

Quinn Lue

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EDUCATION

Montana State University

BS, Computer Science, Minor in Math

3.76 GPA

Aug. 2023 – May 2027

RESEARCH

Long-Context Compression with Pretrained Encoders

December 2025 – Present

- Designed a read-only external memory module (MPNet encoder + cross-attention retrieval into frozen Pythia models) to compress long-context inputs for decoder-only LMs
- Recovered 73.9% of the performance lost under naive truncation on TinyStories, reducing loss from 3.43 to 2.15 (full-context baseline: 1.70) at 12:1 context compression with only 16 visible autoregressive tokens.
- Performed context-corruption ablations, observing near-monotonic increases in loss proportional to corruption severity

Scrabble AI

November 2024 – January 2025

- Developed a novel stochastic tree-search algorithm to approximate move expected value under rack and board uncertainty
- Achieved an average EV error of 5.63% compared to the frontier Scrabble AI, Quackle

PROJECTS

Deep Learning Library (DLX)

June 2025 – September 2025

- Created a custom deep learning framework in Python from scratch with a NumPy/CuPy backend
- Designed a reverse-mode autograd engine and optimizer stack supporting mixed precision, gradient accumulation, checkpointing, and partially frozen parameter graphs
- Defined core tensor operations with gradient propagation under broadcasting, reductions, and device transfers
- Developed a suite of tests, validating FP32 gradients and forward passes with a tolerance of 1e-8

Independent LLM Pretraining & Finetuning

September 2025 – November 2025

- Pretrained a 250m parameter GPT-2 style Large Language Model (LLM) (8.8b tokens, 1x A40 48 GB, 40h) on OpenWebText2 to achieve a final perplexity of 18.17
- Implemented a custom KV-Cache into DLX for efficient inference
- Built custom training infrastructure including learning-rate schedules, data loading for stable training convergence

EXPERIENCE

Undergraduate Research Assistant

May 2025 – October 2025

Montana State University

Bozeman, MT

- Developed an LLM-based sentence classification pipeline for a social science risk and crisis communication framework, working from expert-defined labeling criteria
- Validated predictions against held-out human annotations, finding bias toward positive sentence framing classifications, suggesting a task specific implicit bias in small Llama models

TECHNICAL SKILLS

Languages: Python

Machine Learning: PyTorch, Deep learning, Natural Language Processing (NLP), RAG, representation learning, Transformers, Generative AI, Multimodal Models

Tooling: HuggingFace, Weights & Biases, Git, Linux